

Product Laws of Exponents

Teacher's Edition

The Period

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Must Whole-number exponents with rational bases. Build each law from expansion, then apply it to rewrite an expression and to evaluate one.

Must not No zero exponent yet — that is 3.02 — and no negative exponents until 3.04. No variables until the Thread A days.

Target I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

MTR **MTR.5.1** — Notes II · **MTR.3.1** — Boards 1, 3, 4 and 7 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–3	5 min
Notes	4–7	14 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	10 min
Independent Set	34	6 min
IXL	35	5 min
Total		53 min

Say these three things out loud today

1. Same base, add the exponents. Different bases, and the law does not apply at all.
2. An exponent outside parentheses reaches every factor inside — the number as well as the letter.
3. A power of a power multiplies; a product of powers adds. Say which one you are using before you write.

Slide by Slide

1. **Product Laws of Exponents.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite: a rational base is squared by squaring numerator and denominator.
3. **Warm-Up — Answers.** Reveal. 1. 9 2. 6 and 7 3. $x = 6$ 4. $\frac{4}{9}$
4. **Notes — Copy the three words.** Ask: what does 2 to the 3 mean? Insist on the answer 'three 2s multiplied', not 'six'.
5. **Notes — Where the rule comes from.** We are not adding the bases and not multiplying the exponents. We are counting factors.
6. **Notes.** Expand $(2 \cdot 5)^3$ as three copies of $(2 \cdot 5)$, regroup with the commutative and associative properties, and read off $2^3 \cdot 5^3$.
7. **Notes.** Product of powers: add. Power of a power: multiply. Both come from counting factors, so count if you forget.
8. **Example 1 — Two ways to the same value.** SPOKEN ONLY.
9. **Example 1 — Worked both ways.** Left column: the law, one exponent, then the value.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who multiplied the exponents ($2^{12} = 4096$) or multiplied the bases (4^7).
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A negative base and a power of a power.** SPOKEN ONLY.
13. **Example 2 — Worked.** Multiply the exponents: six factors of -2 .
14. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the added-exponent error ($(-3)^5 = -243$, a negative answer) and the dropped sign (a student who writes 3^6 gets 729 too — ask them to write the base with its sign).
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 16-17. **Board 1.** $5^2 \cdot 5^3 \rightarrow 5^5$ 25^5 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; 5^6 — multiplied the exponents [confused with power of a power]
- 18-19. **Board 2.** A rectangle is 2^4 centimeters long and 2^2 centimeters wide. Its perimeter is 40 centimeters. What is the area of the rectangle, in square centimeters? $\rightarrow 2^6 = 64$ square centimeters The perimeter of 40 centimeters is not needed. 40 — the perimeter was copied as the answer [the unneeded figure was used]; 20 — the sides were added, not multiplied; 256 — the exponents were multiplied (2^8) [confused with power of a power]; 4096 — the bases were multiplied (4^6)
- 20-21. **Board 3.** $(3 \cdot 10)^3 \rightarrow 27,000$ 90 — multiplied the base by the exponent [misconception: exponent treated as a factor]; 3,000 — applied the exponent to one factor only
- 22-23. **Board 4.** $(3^2)^4 \rightarrow 3^8$ 3^6 — added the exponents [confused with product of powers]; 9^8 — squared the base as well
- 24-25. **Board 5.** $4^3 \cdot 4^2$ Which expression is equivalent? $\rightarrow D - 4^5$ (A) multiplied the bases as well as adding the exponents [7.NSO.1.1 misconception: operating on the bases]; (B) added the bases and added the exponents [7.NSO.1.1 misconception: operating

on the bases]; (C) multiplied the exponents — the power-of-a-power move applied to a product [7.NSO.1.1 misconception, B1G-M] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

26-27. Board 6. $(2^4)^3$ Which expression is equivalent? → **C — 2^{12}** (A) added the exponents — the product-of-powers move applied to a power of a power [7.NSO.1.1 misconception, B1G-M]; (B) raised the base to the outer exponent as well [7.NSO.1.1 misconception: operating on the bases]; (D) raised the inner exponent to the outer one, $4^3 = 64$ [misconception: the outer exponent applied to the exponent, not to the power] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

28-29. Board 7. $(-3)^2 \cdot (-3)^1$ → **-27** 27 — the sign was dropped when the odd power was evaluated [7.NSO.1.1 clarification: rational-number bases include negatives]

30-31. Board 8. $(\frac{1}{2})^2 \cdot (\frac{1}{2})^3$ → **$\frac{1}{32}$** 1/64 — multiplied the exponents, $(1/2)^6$ [confused with power of a power]; 5/2 — the exponent 5 written over the 2; 1/10 — the denominator multiplied by 5 instead of raised to the 5th

32-33. Board 9. Aiyanna says she can use the power of a product law to evaluate $(9 + (-3))^3$. Is she right? → **No — $(9 + (-3))^3 = 6^3 = 216$** 702 — distributed the exponent over the sum [7.NSO.1.1: the laws apply to products and quotients, never to sums]

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 25^5 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]

Board 2. 40 — the perimeter was copied as the answer [the unneeded figure was used]

Board 3. 90 — multiplied the base by the exponent [misconception: exponent treated as a factor]

Board 4. 3^6 — added the exponents [confused with product of powers]

Board 5. multiplied the bases as well as adding the exponents [7.NSO.1.1 misconception: operating on the bases]

Board 6. added the exponents — the product-of-powers move applied to a power of a power [7.NSO.1.1 misconception, B1G-M]

Board 7. 27 — the sign was dropped when the odd power was evaluated [7.NSO.1.1 clarification: rational-number bases include negatives]

Board 8. 1/64 — multiplied the exponents, $(1/2)^6$ [confused with power of a power]

Board 9. 702 — distributed the exponent over the sum [7.NSO.1.1: the laws apply to products and quotients, never to sums]

Materials Whiteboards and markers. Calculators are allowed on everything and are useful for the big values in the bank (8^8 , 35^4); the laws themselves are done by hand.